

- **Wireshark (earlier called Ethereal):** A GUI-based packet-capturing and protocol forensic tool.
- **Snort:** An open source network intrusion detection system.
- **Nmap:** A port-scanning and fingerprinting network utility.
- **Bro IDS:** An IDS and network analysis tool.
- **URL Snooper:** Locates the addresses of audio and video files to enable recording them.
- **Iftop:** Displays the usage of bandwidth in the network.
- **EtherApe:** GUI-based tool for monitoring network traffic and bandwidth usage in real time.
- **Bit-Twist:** Ethernet packet generator.
- **Pirni:** Network security tool used with jailbroken iOS devices.
- **Firesheep:** An extension for the Mozilla Firefox Web browser. It intercepts the unencrypted cookies from different websites. It is also used for session hijacking and network vulnerabilities.
- **Suricata:** A network intrusion analysis and prevention platform.
- **WhatPulse:** A statistical measurement tool (input, network, uptime, etc) in the network.
- **Xplico:** A network forensics analysis tool (NFAT).

Snort

Snort is an open source tool developed by Sourcefire and written in C. It is used as a network intrusion prevention as well as a network intrusion detection system. It is an excellent combination of signature, protocol and anomaly-based inspection. Snort's user statistics are impressive with millions of downloads and nearly 400,000 registered users. It is not only an IDS but is also used as an IPS (Intrusion Prevention System) to avoid any unwanted activity or unauthorised access to resources. Snort can easily implement protocol analysis and content investigation with a number of other features. The excellent features of the tool include detection of a variety of attacks and probes such as buffer overflows, stealth port scans, CGI attacks, SMB probes, OS fingerprinting attempts and many others.

The main configuration file is `/etc/snort/snort.conf`. In this, the actual information about the network or system under investigation is specified. All values and parameters are commented in the file so that the changes can be made very easily.

Other free intrusion detection systems (IDS)

Other free intrusion detection systems are given below:

- ACARM-ng
- AIDE
- Bro NIDS
- OSSEC HIDS
- Prelude Hybrid IDS
- Samhain
- Suricata

Different modes for Snort network forensics

Snort can be configured in three different modes.

Sniffer mode: Sniffer mode is used to read the packets of the network and display them to the user in a continuous stream

on the system.

Packet logger mode: Packet logger mode is used to log the packets to the disk.

Network IDS mode: Network IDS mode is associated with configurable parameters that allow the tool to analyse network traffic by matching it against specific user-defined rules. Finally, it performs several actions based on what the tool investigates at the time of execution.

Sniffer mode

To view only the TCP/IP packet headers to the screen, the following command is used:

```
./snort -v
```

This will execute Snort and show the IP and TCP/UDP/ICMP headers.

To display the application data in transit, the following command is used:

```
./snort -vd
```

With this, Snort displays the packet data as well as the headers.

To view a detailed analysis or description, showing the data link layer headers, use the following command:

```
./snort -vde
```

This command, with different switches, can also be executed as:

```
./snort -d -v -e
```

Packet logger mode

To record the packets to a disk, we need to specify the directory for logging. Snort will automatically execute the packet logger mode:

```
./snort -dev -l ./log
```

Running in this mode, Snort will collect every data packet that it encounters and place it in a directory hierarchy that is based upon the IP address of one of the hosts in the datagram.

If you specify a switch `-I'`, you will see that Snort uses the address of the remote computer as the directory in which it places packets. In some scenarios, it makes use of the local host address. For the logging of packets relative to the home network, you have to specify this network with the following command:

```
./snort -dev -l ./log -h 192.168.1.0/24
```

This command and rule will specify that you want to view the data link and TCP/IP headers as well as application data